

Automated Diagnosis of Cancer Cells Using Supervised Machine Learning

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Introduction

Machine learning and AI predictive models have been in the healthcare field for decades, one of the first dating back to the 1970's where an AI program was designed to diagnose bacterial infections and from that, recommend the antibiotics to treat said bacteria (Hirani et al., 2024). As technology advances, so do predictive models. Using machine learning such as Logistic Regression, Random Forest, SVM, Naive Bayes, and many others has allowed for progress in the predictive model realm of healthcare. By discovering how to better train the models and obtaining robust datasets the path to more efficient predictive models is rising fast. However, since each dataset is different and there are many types of model it is important to find the right match for the dataset being worked with.

Due to the importance of finding the best fit model this project takes the breast cancer dataset and tests four different model approaches (Logistic regression, Random Forest, SVM, and Naive Bayes) in attempts to train a successful predictive model for identifying malignant vs benign masses. In addition, the project aims to compare model classification performance to determine the best and most reliable machine learning modeling method.

Dataset

For this project the Breast Cancer Wisconsin (Diagnostic) Data Set from the Scikit-learn library was used. The dataset consists of 569 masses. Each sample contains a target feature, along with 10 core features (radius, texture, perimeter, area, smoothness, compactness, concavity, concave point, symmetry, and fractal dimension). These 10 core features are analyzed from three different perspectives (mean, error, and worst) to create 30 total features, which when combined with the initial target feature, totals for 31 features per sample.

Data Cleaning and Pre-processing

The dataset was evaluated for missingness and duplicate rows; none were found. Next when evaluating the target class, malignant vs benign, bias was observed: 62% benign vs 37% malignant. In order to obtain a meaningful sample to train the models we must adjust the dataset to remove any bias. **S**ynthetic **M**odel **O**versampling **T**Echnique or SMOTE was utilized to synthetically over

sample the malignant section to match the benign sample. This technique will synthesize a new datum based off the average distance between two random existing points. This allows for synthetic data creation without expanding the maximum or minimum values of the dataset. (Malkin, 2022)

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is important when working with a new dataset. Approaching a problem with an agenda allows for better statistical observation and analysis. A correlation heatmap matrix is a great way to initiate EDA. This shows positive and negative correlations for every variable against every variable.

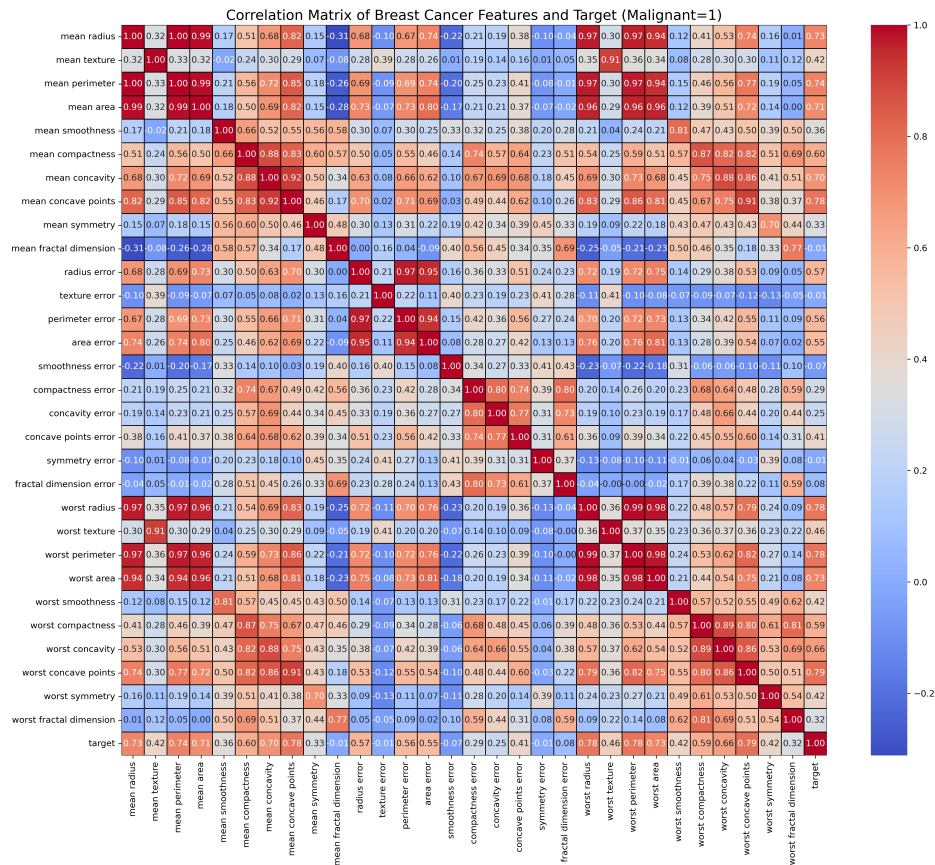


Figure 1: Correlation heatmap of features.

In Figure 1, we can observe that almost every variable has a positive correlation with the target value (malignant = 1), with worst concave points, worst perimeter, worst radius, and mean concave points having the highest correlation. However, just because they have a high correlation coefficient does not imply causation, other factors can be at play that we need to pay attention to. For example looking at mean area with a coefficient of 0.71 to target would imply as mean area increases so does the likelihood of the mass being malignant. But mean area to mean radius coefficient is 0.99 and mathematically radius and area are linked. Also mean radius to target has a higher coefficient of 0.73 than 0.71 from mean area to target. Does this mean that area is truly related to

target or only related to target due to its link to radius? This question is why correlation does not automatically imply causation on it's own. This is just one example we can see how correlation does not equal causation within the dataset, but there are more. This being a known problem the hope of EDA and correlation matrices is to find several factors that correlate in the larger picture, then apply advanced processing techniques to find the true relationship between the variables and the target. For this project we used PCA to help decrease the noise in the dataset.

Feature Engineering

Noise is inherent in any dataset. Our world is filled with data, it is difficult to narrow down which variables are correlative vs causal links. The dataset used has 30 features compared to the target feature. With so many relationships it is helpful to be able to reduce the dimensions of our dataset. Luckily with **Principal Component Analysis** this task becomes easier.

Principal component analysis reforms the variables in a dataset in attempts to find what groups of variables best explains the dataset without losing too much information (IBM, 2025). By applying PCA to our 30 variables we can identify which variables have the highest weight in aligning with the target variable. The best way to determine how many principal components to train the models with is by identifying the best variance utilization with a Scree plot.

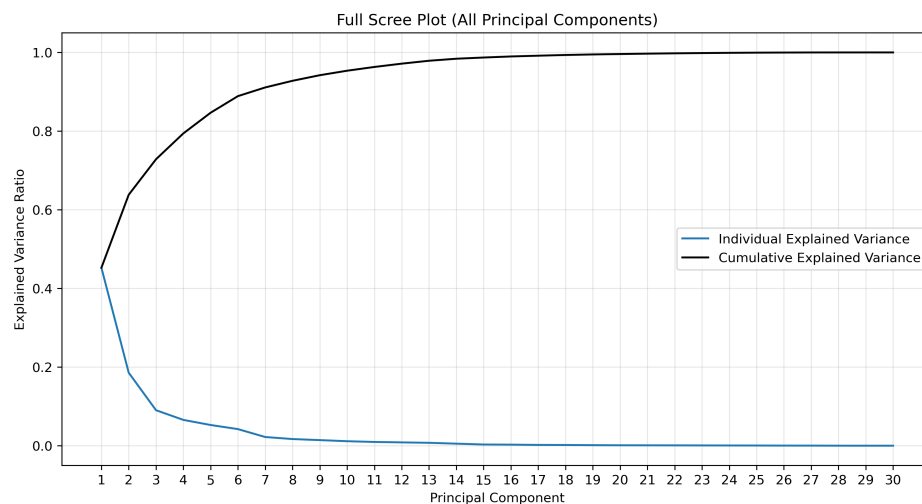


Figure 2: Scree plot of all principal components.

In the Scree plot from Figure 2 displays the 30 principal components ranked by the highest individual explained variance. After 7 principal components, roughly 95 percent of the dataset's variance is accounted for or explained. With this analysis we can reduce our 30 variables down to the 7 principle components and train the models, reducing the computation time without losing too much data (IBM, 2025). However, just selecting how many principal components to use is not the only battle. There are many ways to train a model, each with their own application. To discover which model is best we used four different machine learning model architectures: Logistic Regression, Random Forest, Support Vector Classifier, and Naive Bayes.

Methods

Logistic Regression

The first model we implemented was Logistic Regression. Logistic Regression assumes a linear relationship between the independent variables (features) and the log odds of the dependent variable (target). This model uses the sigmoid function to calculate the probability that an observation belongs to a given class. Given more formally, the model computes a linear combination of input features given by

$$z = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p.$$

The linear combination, z , is then passed through the sigmoid function, defined as

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The sigmoid function can map z into a probability value between 0 and 1. These probabilities are then used to decide the class label following a threshold value, typically given as

$$\text{Prediction} = \begin{cases} \text{Class 1 (malignant)} & \text{if } \sigma(z) > 0.5, \\ \text{Class 0 (benign)} & \text{if } \sigma(z) < 0.5. \end{cases}$$

Random Forest

The second model we implemented was Random Forest. Random Forest is an ensemble method which utilizes many individual decision trees to decide based on majority voting (in the case of classification). The internals of this model rely on bagging and feature randomness, which involves generating multiple bootstrap samples of the training data and training separate decision trees on each subset. In addition, random subsets of features are considered at each split. This combination of bootstrapped data and feature level randomness allows the model to produce diverse trees which are decorrelated and robust. Once training is complete, the predictions from all trees are aggregated using majority voting, where the class chosen by the greatest number of trees is assigned as the final classification.

Support Vector Classifier

The third model implemented was Support Vector Classifier (SVC). A Support Vector Classifier aims to find the optimal separating hyperplane which maximizes the margin between classes. By maximizing the margin, the model can generalize better to unseen data. Another important feature of the SVC model is its kernel, which are functions that map data into a higher dimensional input feature space, allowing for a linear separating hyperplane to be found. For this project, we will be utilizing the radial basis function (RBF kernel), which allows the model to learn nonlinear and complex decision boundaries.

Naive Bayes

The last model implemented was Naive Bayes. Naive Bayes is a probabilistic classifier that utilizes Bayes' Theorem to estimate the probability of an observation belonging to a particular class. This probability is calculated by combining the prior probability of each class with the likelihood of the observed features given that class. To further simplify this calculation, the model assumes that all features are conditionally independent given the class label. Formally, the joint likelihood of the features becomes

$$P(x_1, x_2, \dots, x_n | y) = P(x_1 | y) \cdot P(x_2 | y) \dots P(x_n | y) = \prod_{i=1}^n P(x_i | y).$$

Bayes' Theorem for classification can then be written as

$$P(y | x_1, \dots, x_n) = \frac{P(x_1, \dots, x_n | y) P(y)}{P(x_1, \dots, x_n)} = \frac{\prod_{i=1}^n P(x_i | y) \cdot P(y)}{\prod_{i=1}^n P(x_i)},$$

$$P(y | x_1, \dots, x_n) \propto \prod_{i=1}^n P(x_i | y) \cdot P(y).$$

In our application where features are continuous, we use the Gaussian Naive Bayes model variant, which assumes that each feature x_i follows a Gaussian (normal) distribution within each class. This can be modeled by

$$x_i | y \sim \mathcal{N}(\mu_{iy}, \sigma_{iy}^2).$$

This assumption transforms the likelihood into

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_{iy}^2}} \exp\left(-\frac{(x_i - \mu_{iy})^2}{2\sigma_{iy}^2}\right).$$

The model then calculates the posterior for each class y and classifies the observation based on highest probability, given by

$$\hat{y} = \arg \max_y \left(\prod_{i=1}^n P(x_i | y) \cdot P(y) \right).$$

Together, these aspects of the Naive Bayes model make for a computationally efficient classifier which performs well for high dimensional data.

Hyperparameter Tuning

To further improve model performance, hyperparameter tuning was performed for the Random Forest and SVC models. A grid search was conducted over key hyperparameters for each model, with optimal values selected based on 5-fold cross validation accuracy. Once the optimal hyperparameters were identified, the tuned models were re-trained on the training data and compared to their respective baselines to assess improvements or declines in classification performance.

For the Random Forest model, the tuning grid included the following hyperparameters:

- **n_estimators**: number of trees in the forest
- **max_depth**: maximum depth of each decision tree
- **max_features**: number of features considered at each split

For the SVC model, the grid search included:

- **C**: regularization parameter controlling the penalty for misclassification
- **gamma**: parameter controlling the influence of individual training samples in the RBF kernel

Evaluation Metrics and Methods

To evaluate the performance of the models on the testing set, the following key metrics were used to assess model classification performance:

- **Accuracy**: The proportion of cancer cells correctly classified. Higher accuracy values indicate fewer classification errors across both classes.
- **Precision**: The proportion of samples predicted to be malignant that are truly malignant. Higher precision reduces false positives, ensuring that benign cases are not incorrectly classified as malignant.
- **Recall**: The proportion of truly malignant samples that the model successfully identifies. Higher recall reduces false negatives, helping ensure that malignant cases are not missed.
- **F1 Score**: The harmonic mean of precision and recall.

To complement these metrics, additional evaluation methods were implemented to provide a more comprehensive understanding of model performance and behavior. These methods include:

- **Confusion Matrix**: The first evaluation method used was the confusion matrix, generated for each model using the test set. By observing the distribution of true positives, true negatives, false positives, and false negatives, we gain deeper insights into model behavior by examining both correct classifications and error patterns.

- **K-Fold Cross Validation:** The second evaluation method used was 5-fold cross validation, implemented to assess each model’s stability and variability across different subsets of the training data. This technique evaluates the consistency of a model’s performance rather than relying on a single train-test split, providing a more robust and less biased measure of generalizability.
- **Decision Boundary Visualization:** The final evaluation method used was the visualization of decision boundaries. By projecting the data onto the top two principal components (PC1 and PC2) from the PCA transformation, we can visualize how each model separates malignant and benign samples in a reduced two-dimensional feature space. Comparing these boundaries provides insight into model structure, linearity, and overall complexity.

Results and Discussion

Training Results

Across the four models, Logistic Regression and SVC demonstrated the highest overall performance, each achieving an accuracy of 0.973. Random Forest followed with an accuracy of 0.938, while Naive Bayes performed the weakest at 0.929. In terms of precision, Logistic Regression ranked the highest, followed by SVC, Random Forest, and Naive Bayes respectively. For recall, SVC performed best, slightly surpassing Logistic Regression, with Random Forest and Naive Bayes again following behind. The pattern remained similar for the F1 score, where SVC achieved the top value, closely followed by Logistic Regression, then Random Forest, and Naive Bayes last.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.973684	0.985915	0.972222	0.979021
Random Forest	0.938596	0.957746	0.944444	0.951049
SVC (RBF)	0.973684	0.972603	0.986111	0.979310
Naive Bayes	0.929825	0.957143	0.930556	0.943662

Table 1: Model performance comparison using accuracy, precision, recall, and F1 score.

Overall, these results indicate that SVC and Logistic Regression offer the most balanced and reliable performance, excelling across nearly all evaluation metrics. Random Forest provides solid but slightly less consistent performance, and Naive Bayes, under performs relative to the other models.

Confusion Matrices

Looking at the confusion matrices, we can observe that models with higher accuracy, such as Logistic Regression and SVC, have more predictions along the main diagonal, which indicate correct classifications, and fewer off diagonal predictions, indicating less false positive and false negative classifications. On the other hand, the models with lower accuracy, such as Random Forest and Naive Bayes have more predictions on the off diagonals, revealing more classification errors.

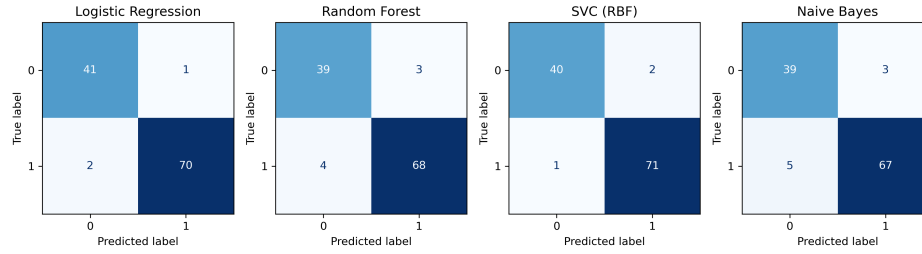


Figure 3: Confusion matrices for all four baseline models.

K-Fold Cross Validation

To visualize accuracy scores across folds, a box plot was used to compare model consistency and average performance. From the boxplot, we can observe that Logistic Regression and SVC achieve higher median accuracy and tighter interquartile ranges, indicating low variation and stable performance across folds. These behaviors suggest that the Logistic Regression and SVC models are generalizing well. Contrarily, the Random Forest and Naive Bayes models exhibit lower median accuracy and greater variability in their interquartile ranges, suggesting unstable performance across folds.

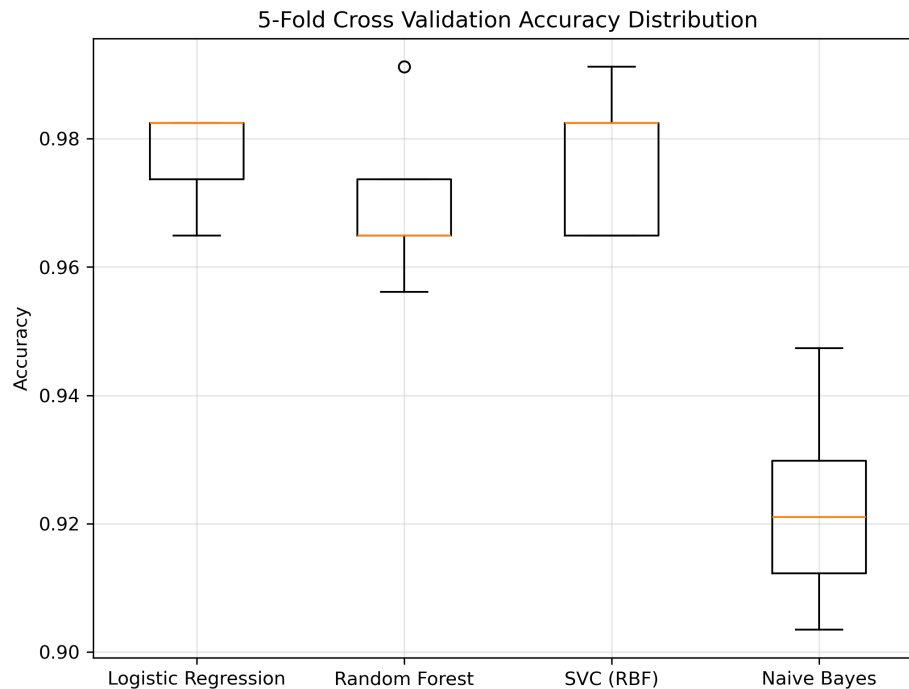


Figure 4: Boxplot of cross validation accuracies for each model.

Decision Boundary Visualizations

From the decision boundary plots, we can observe the structure of class partitions for each model.

- Logistic Regression displays a linear decision boundary, reflecting the model's assumption

of a linear relationship between features and class labels.

- Random Forest produces piecewise, irregular partitioned regions that arise from the model's threshold-based splitting rules.
- SVC generates a smooth, nonlinear decision boundary formed by maximizing the margin between classes using the RBF kernel.
- Naïve Bayes yields elliptical nonlinear boundaries due to its use of Gaussian likelihood estimations for each feature.

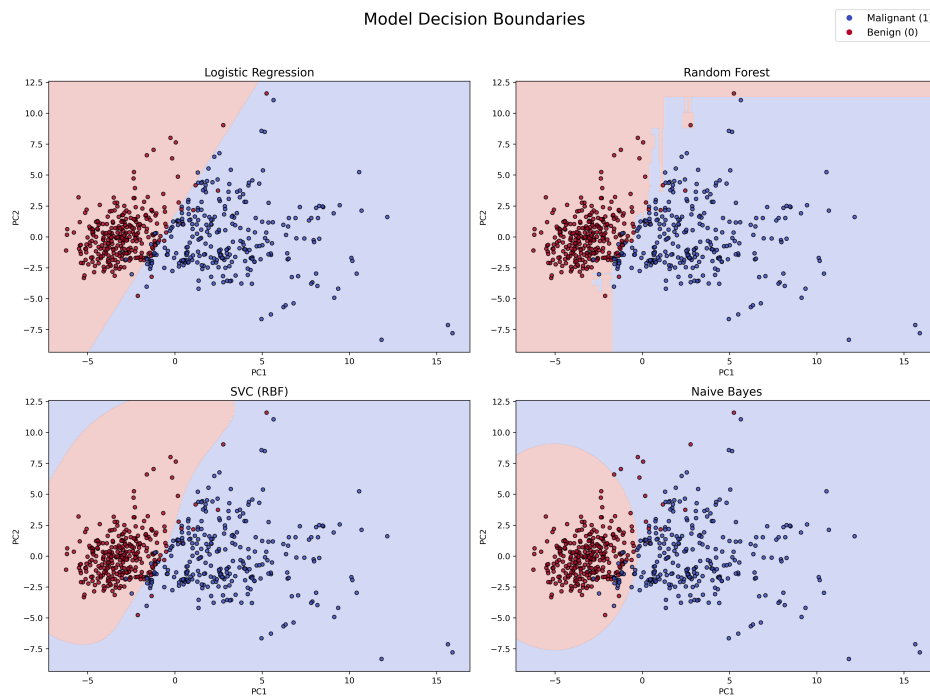


Figure 5: 2-dimensional decision boundaries for each model.

By visually comparing these decision boundaries, we gain an interpretable understanding of model structure, complexity, and class separation behavior.

Tuned Models

Using GridSearchCV, the optimal hyperparameters identified for the Random Forest model were:

- `n_estimators = 300`
- `max_depth = None`
- `max_features = sqrt`

This tuned model achieved an accuracy of 0.947, higher than the baseline Random Forest model accuracy of 0.938. In addition, this tuned model also demonstrated higher values across all the

other performance metrics (precision, recall, F1 score), indicating that a more complex model utilizing more estimators and unrestricted tree depth contributed to better classification performance.

For the SVC model, GridSearchCV identified the following optimal hyperparameters:

- $C = 10$
- $\text{gamma} = \text{scale}$

After tuning, the SVC model achieved lower performance across all four metrics compared to its baseline. These results suggest that the baseline SVC model was already optimal and therefore tuning reduced model classification performance.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.947368	0.971429	0.944444	0.957746
SVC (RBF)	0.956140	0.946667	0.986111	0.965986

Table 2: Tuned model performance comparison.

Thus, we can conclude that hyperparameter tuning was beneficial for Random Forest as the model was able to improve its performance, but not significantly important for models which are already near optimal.

Conclusion

The goal of the project was to compare multiple machine learning models to evaluate their performance on the task of cancer cell classification. By training several baseline models and conducting hyperparameter tuning, we aimed to identify which models achieved the best classification performance and demonstrated consistent, reliable behavior under a standardized evaluation framework.

Across the models we tested, we can conclude that Logistic Regression and SVC performed the best, with each model achieving an accuracy of 0.973 on the testing set. Random Forest and Naive Bayes also performed well, each achieving slightly lower baseline accuracy scores of 0.938 and 0.929 respectively.

From our hyperparameter tuning framework, we found that the model optimization process yielded mixed results, with Random Forest gaining classification performance across all metrics, while SVC experienced a decrease in performance relative to their baselines.

Overall, our project successfully developed and evaluated a set of machine learning models with varying architectures and levels of complexity. These results illustrate how machine learning can be utilized in biomedical applications for diagnosis decision-making.

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